

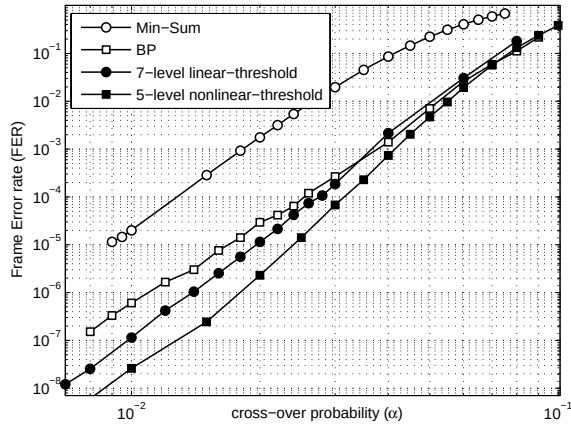


Fig. 3. FER results on the $n = 155$, $R = 0.4$, Tanner code

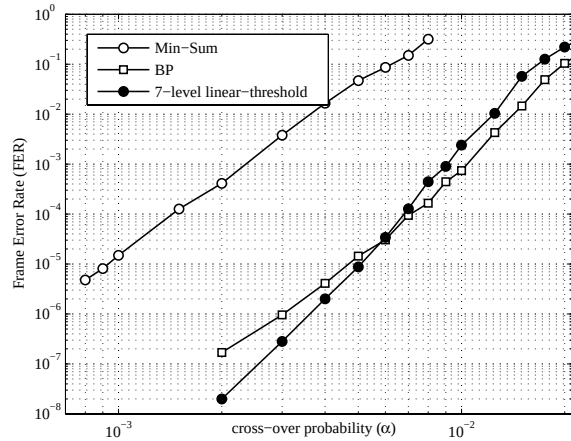


Fig. 4. FER results on the $n = 768$, $R = 0.75$, Quasicyclic code

decoder guarantees correction of all error patterns up to 5 errors. An interesting point to note is that the highly complex linear programming (LP) decoding fails to correct all 5-errors on the Tanner code, whereas the 3-bit decoder is able to, which illustrates the power of multilevel decoding.

VI. DISCUSSION AND CONCLUSIONS

Multilevel decoding was established as a powerful decoding technique for LDPC codes. From our concluding example, we highlighted the importance of the isolation assumption as a crucial strategy for deriving good rules. Also from the results, it is evident that there is a certain aspect of universality to these decoders; the same 3-bit decoder performs well on a variety of codes. This suggests that deriving good rules based on trapping sets appears to take the structure of the local neighborhood into local computations of messages, which is a tiny step closer in our pursuit to approach maximum-likelihood decoding. Our future work includes deriving bounds and relating the isolation theorem with guaranteed error-correction capability.

ACKNOWLEDGMENT

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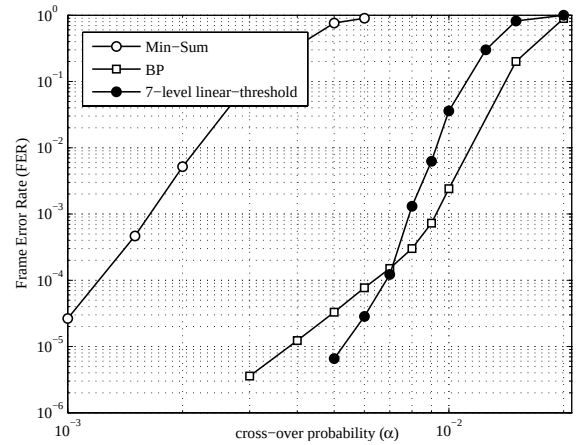


Fig. 5. FER results on the $n = 4085$, $R = 0.82$, MacKay code

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